Recoverable Feature Routing under Distribution Shift: Real-data T4 bridge

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Abstract

The controlled mechanism is translated into a T4-compatible real-image pipeline with four ResNet-18 evidence experts, cached logits and contexts, resumable stages, routing adaptation, and selective activation. Dependency resolution, Python compilation, and CLI construction were verified. No historical full-run result archive was recovered, so this release is code-only and makes no numerical real-data claim.

Historical Manuscript Status

This is the cumulative scientific manuscript at release **v0.6.0**. It includes only experiments completed by 4 October 2026, 20:17 UTC; later findings are intentionally absent.

1. Introduction

Neural classifiers usually prioritize features that are most predictive across their training distribution. That behavior is efficient when feature reliability is stable, but brittle when a dominant cue fails while lower-priority evidence remains informative. The motivating deepfake-detection case is a model that trusts one family of texture, frequency, chromatic, geometric, or localized facial evidence until eyewear, occlusion, compression, or generator shift changes its reliability.

The desired response is not necessarily permanent retraining. A system should reconsider which evidence it trusts for the affected regime while preserving its established behavior elsewhere. This manuscript records a sequence of controlled falsification experiments designed to separate apparent adaptation from actual access to neglected evidence.

Each tagged version is a historical scientific snapshot. Negative results, implementation failures, and abandoned explanations are retained because they determine why the next experiment exists.

2. Related Work

Mixture-of-experts systems use learned gates for input-dependent expert selection [1,2]. MAML and fast-weight systems formalize rapid adaptation [3,4], while test-time adaptation and continual learning expose complementary distribution-shift and stability problems [5,6]. The present sequence isolates a narrower question: after a learned evidence policy becomes unreliable, does useful backup evidence remain both represented and reachable by a bounded adaptation process?

3. Controlled Experimental Setting

3.1 Evidence families Each synthetic image belongs to one of two classes and contains three separately accessible evidence families:

- **A: eye-region evidence.** A nearly deterministic localized marker makes this the dominant cue on clean data.
- **B: lower-face geometry.** A weaker geometric pattern is correlated with the label with probability 0.72.
- **C: chromatic evidence.** A weaker global color bias is correlated with the label with probability 0.66.

Three small convolutional experts receive feature-isolated views. Expert A sees the eye band, expert B sees grayscale lower-face geometry, and expert C sees an opponent-color representation.

The separation is intentional. It allows us to measure whether a model retains genuinely useful backup information after A fails, rather than inferring feature behavior from a monolithic latent representation.

3.2 Held-out intervention The final intervention is a synthetic glasses overlay applied to the eye band. It is not used during slow-router training. The perturbation is designed to invalidate A while leaving B and C largely intact.

The router is meta-trained on other perturbations, including eye-band masking, lower-face masking, desaturation, blur, JPEG compression, and clean inputs. Thus, the system is exposed to the general idea that evidence reliability can change, but it does not learn a direct glasses-specific rule.

3.3 Evaluation protocol The principal runs use five seeds: 11, 22, 33, 44, and 55. Few-shot support sizes are 2, 4, 8, 16, and 32 labeled glasses examples. Support samples are excluded from query evaluation.

The main metrics are shifted-regime accuracy, clean accuracy after adaptation, expert routing weights, routing entropy, and saturation rate. Later experiments add explicit measures of dominant-cue release, backup-cue recovery, routing Jacobian conductance, and initial backup-gradient magnitude.

4. Experiments

4.1 Experiment 1: failure of an overly easy toy world The first toy construction contained RGB, edge, high-pass, and chromatic expert views. It failed as a falsification environment because all fusion strategies achieved 1.000 accuracy on both clean and glasses-corrupted data. The perturbation increased cross-entropy loss, but the edge expert remained sufficiently diagnostic that the router had no incentive to rediscover weak evidence.

This negative result motivated a stricter construction in which the globally dominant feature is explicitly fragile.

4.2 Experiment 2: fragile dominance and few-shot recovery Experiment 2 introduced evidence families A, B, and C. On clean data, the learned global fusion placed approximately 95.7% of routing mass on A. Under glasses, the fixed global system fell to exactly 0.5000 mean accuracy across five seeds.

The backup experts remained useful. Under glasses, B and C independently retained above-chance classification performance, while A collapsed to chance. A support-optimized fusion reference reached 0.6944 ± 0.0302 accuracy at 32 shots.

The adaptive router partially recovered:

Method	K	Glasses accuracy
Global fixed fusion	0	0.5000 ± 0.0000
Adaptive zero-shot	0	0.5008 ± 0.0016
Adaptive few-shot	2	0.5872 ± 0.1052
Adaptive few-shot	8	$\mathbf{0.6302 \pm 0.1064}$
Adaptive few-shot	32	0.6300 ± 0.1061
Support-optimized fusion reference	32	0.6944 ± 0.0302

At 8 shots, mean routing changed from approximately

$$[A, B, C] = [0.904, 0.088, 0.008]$$

before glasses-specific adaptation to

$$[A, B, C] = [0.451, 0.539, 0.010].$$

The mechanism therefore worked in aggregate, but the standard deviation was large. Inspection showed that some seeds escaped from A while others remained almost perfectly locked to it.

4.3 Experiment 3: output anti-collapse is not gradient recoverability Experiment 3 compared ordinary softmax routing with temperature scaling, an epsilon probability floor, logit clipping, and bounded residual routing.

The main result was negative but informative. An epsilon floor eliminated measured output saturation while improving glasses accuracy only to 0.5444 ± 0.0878 at 32 shots. Logit clipping reached 0.5877 ± 0.1074 at 8 shots, similar to ordinary softmax. No anti-collapse transformation consistently eliminated the seed-dependent failure.

This shows that a router can assign nonzero mass to a backup expert without making that expert meaningfully accessible to gradient-based adaptation. For an epsilon-floor transformation,

$$w_i = (1 - E\epsilon)\,\mathrm{softmax}(z)_i + \epsilon,$$

the output probability may be bounded away from zero while the derivative with respect to the underlying logit remains proportional to the already saturated softmax derivative.

The relevant property is therefore not merely whether $w_i > 0$, but whether the adaptation parameters retain sufficient local influence over w_i.

4.4 Experiment 4: independent fast pathways Experiment 4 directly tested this gradient-accessibility hypothesis. All methods within a seed shared the exact same expert bank, slow router, support set, and query set.

A standard slow update and an additive fast-logit update both continue to route through the slow softmax geometry. In contrast, the independent mixture uses

$$p_s = \mathrm{softmax}(z_s), \qquad p_f = \mathrm{softmax}(a),$$

and

$$w = (1 - \alpha)p_s + \alpha p_f.$$

The fast distribution therefore has a routing Jacobian independent of the slow distribution.

The fast-only route produced the strongest shifted-regime recovery:

Method	Best K	Glasses accuracy	Clean accuracy after adaptation
Slow SGD	8	0.6302 ± 0.1064	0.9226
Additive fast logits	8	0.6302 ± 0.1064	0.8668
Fixed independent mixture	32	0.5657 ± 0.0805	1.0000
Learned independent mixture	4	0.6070 ± 0.0954	**0.9974**
Fast-only independent path	32	**0.6986 ± 0.0259**	0.7158

The fast-only route also had markedly lower variance than the slow adaptive router. Its initial routing conductance was 0.4714, compared with approximately 0.14 on average for the slow route and nearly zero on the most strongly locked seeds.

These results support two conclusions.

First, a fresh parameter vector is not sufficient by itself: if it is inserted before the same saturated softmax, adaptation can still fail.

Second, a computationally independent fast distribution restores gradient access, but indiscriminate application of the fast policy creates forgetting on the clean regime.

This produces a clear stability-plasticity frontier.

4.5 Experiment 5: support-keyed selective activation Experiment 5 closes the stability-plasticity loop exposed by Experiment 4. The expert bank, slow router, support set, query set, and adapted fast policy are held fixed across activation methods. The only experimental variable is **when** the fast path is invoked.

The final route is

$$w(x) = (1 - \alpha(x))p_s(x) + \alpha(x)p_f.$$

Five seeds are evaluated with $K \in \{2, 4, 8, 16, 32\}$. A balanced clean reference memory of 128 samples is drawn from the validation distribution and is never taken from the clean test set.

The prototype switch uses a frozen context embedding $\phi(x)$:

$$s(x) = \cos(\phi(x), \mu_{\text{support}}) - \cos(\phi(x), \mu_{\text{clean}}).$$

The memory switch instead trains a small linear environment classifier on frozen context features, using glasses support examples as the new-regime class and the clean reference memory as the old-regime class. It never receives fake/real targets.

The principal joint metric is

$$H = \frac{2A_{\text{glasses}}A_{\text{clean}}}{A_{\text{glasses}} + A_{\text{clean}}}.$$

The results are:

Method	Best K by joint score	Glasses accuracy	Clean accuracy	Harmonic score
Slow SGD	8	0.6302 ± 0.1064	0.9226 ± 0.1081	0.7367
Fast only	4	0.6669 ± 0.0701	0.8018 ± 0.1270	0.7179
Global slow-fast mixture	8	0.5925 ± 0.0836	0.9674 ± 0.0652	0.7294
Prototype switch	32	0.5636 ± 0.0790	1.0000 ± 0.0000	0.7177
Memory switch	**32**	0.6986 ± 0.0259	1.0000 ± 0.0000	**0.8223**
Environment oracle	32	0.6986 ± 0.0259	1.0000 ± 0.0000	0.8223

The memory switch exactly matches the environment-oracle accuracies at every tested support size in this synthetic setting. At $K = 32$,

$$\bar{\alpha}_{\text{glasses}} = 1.000, \qquad \bar{\alpha}_{\text{clean}} \approx 0.0488,$$

with a glasses activation rate of 1.000 and a clean false-activation rate of 0.000.

This means that the model preserves the full recovery of the independent fast policy on shifted inputs while retaining the unadapted slow behavior on the clean test distribution. In the controlled toy setting, the stability-plasticity trade-off from Experiment 4 is therefore almost completely removed by context-dependent fast-path activation.

The prototype switch is also informative. Its thresholded regime decision is nearly perfect—at $K = 32$, the glasses activation rate is 1.000 and the clean false-activation rate is 0.000—but its mean mixture coefficients remain close to 0.5:

$$\bar{\alpha}_{\text{glasses}} \approx 0.5105, \qquad \bar{\alpha}_{\text{clean}} \approx 0.4939.$$

Thus, the representation contains enough information to separate the regimes, but the continuous prototype-to-mixture calibration is too weak. This is a calibration failure rather than a failure of regime discrimination.

These results support a three-part mechanism:

1. backup evidence must remain represented;
2. an adaptation path to that evidence must remain locally accessible;
3. the fast path should be activated conditionally, rather than globally.

The result is a mechanism demonstration, not evidence that the same separation will be easy on natural deepfakes. The synthetic glasses intervention is visually regular, and the memory switch receives a small clean reference set. A realistic study must test whether similarly clean regime separation survives natural eyewear, generator shift, compression, pose, and multiple simultaneous nuisance factors.

4.6 Experiment 6: real-data deepfake bridge Experiment 6 translates the mechanism into a T4-oriented real-image pipeline. Four ResNet-18 experts receive RGB, high-pass, chromatic, and structural views. Experts train sequentially to respect accelerator memory limits. Frozen logits and context embeddings are cached so that routing experiments do not repeatedly execute the full backbone stack. The implementation includes procedural eyewear, slow and fast routing, selective activation, multi-seed execution, resumable stages, and result packaging.

Component	Scientific role
RGB expert	ordinary spatial appearance evidence
High-pass expert	residual and frequency-sensitive evidence
Chromatic expert	opponent-color and channel consistency
Structural expert	geometry-oriented evidence
Cached logits/context	decouple representation learning from routing studies
Slow/fast router	test recoverability and selective activation
Resume checkpoints	make the study feasible on Colab T4

During archival reconstruction, dependency resolution, Python compilation, and CLI construction passed. No historical full-run result archive was recovered and no replacement full dataset run was performed. This release therefore establishes executable infrastructure only. It does not support a real-data performance or transfer claim.

Interpretation at This Milestone

The controlled mechanism now has a runnable real-data bridge, but transfer remains untested. Completed real-data runs, dataset provenance, checkpoints, and condition-stratified metrics are required before any empirical claim.

Reason for the Next Stage

No further stage is claimed complete. The next valid step is a fully recorded real-data run and condition-stratified analysis.

Limitations and Threats to Validity

The controlled worlds are binary, visually regular, and explicitly factorized. The support sets are labeled, only five principal seeds are used after the first stages, and the synthetic glasses intervention may be easier to recognize than natural distribution shifts. Best-operating-point comparisons may benefit from multiple testing. Experiment 0 lacks raw result artifacts, while Experiment 6 lacks completed results. Executable smoke tests establish software integrity, not reproduction of full-scale scientific outcomes.

Reproducibility and Provenance

The accompanying repository separates byte-preserved historical source and results from repaired runnable snapshots. Each runnable stage provides a `uv` environment, exact entrypoint, bounded smoke test, and validation record. Reconstructed proposal, implementation, results, and analysis commits preserve why the sequence changed. Numerical claims come from stored outputs unless explicitly marked reported-only or results-missing.

Disclosure of AI Assistance

ChatGPT collaborated in experimental ideation, implementation drafting, debugging, interpretation, manuscript preparation, and archival reconstruction. Git commits credit ChatGPT through co-author trailers. The exact ChatGPT model identifiers and reasoning-effort settings used during the original 4 October 2026 experiment conversations were not exposed in the recovered metadata and have not been inferred. Mykyta Siryk remains responsible for reviewing the code, evidence, claims, and manuscript.

References

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